



# S1A THRU S1M

## 1.0 AMP. Surface Mount Rectifiers



Voltage Range  
50 to 1000 Volts  
Current  
1.0 Ampere

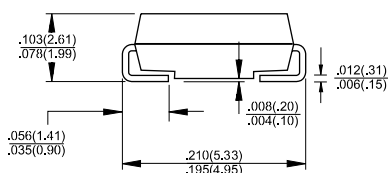
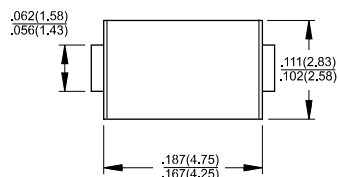
### Features

- ✧ For surface mounted application
- ✧ Glass passivated junction chip.
- ✧ Low forward voltage drop
- ✧ High current capability
- ✧ Easy pick and place
- ✧ High surge current capability
- ✧ Plastic material used carries Underwriters Laboratory Classification 94V-O
- ✧ High temperature soldering:  
250°C / 10 seconds at terminals

### Mechanical Data

- ✧ Case: Molded plastic
- ✧ Terminals: Solder plated
- ✧ Polarity: Indicated by cathode band
- ✧ Packaging: 12mm tape per EIA STD RS-481
- ✧ Weight: 0.064 gram

### SMA/DO-214AC



Dimensions in inches and (millimeters)

### Maximum Ratings and Electrical Characteristics

Rating at 25°C ambient temperature unless otherwise specified.

Single phase, half wave, 60 Hz, resistive or inductive load.

For capacitive load, derate current by 20%

| Type Number                                                                                               | S1A         | S1B | S1D | S1G | S1J | S1K | S1M  | Units    |
|-----------------------------------------------------------------------------------------------------------|-------------|-----|-----|-----|-----|-----|------|----------|
| Maximum Recurrent Peak Reverse Voltage                                                                    | 50          | 100 | 200 | 400 | 600 | 800 | 1000 | V        |
| Maximum RMS Voltage                                                                                       | 35          | 70  | 140 | 280 | 420 | 560 | 700  | V        |
| Maximum DC Blocking Voltage                                                                               | 50          | 100 | 200 | 400 | 600 | 800 | 1000 | V        |
| Maximum Average Forward Rectified Current<br>@T <sub>L</sub> =110°C                                       | 1.0         |     |     |     |     |     |      | A        |
| Peak Forward Surge Current, 8.3 ms Single<br>Half Sine-wave Superimposed on Rated Load<br>(JEDEC method ) | 30          |     |     |     |     |     |      | A        |
| Maximum Instantaneous Forward Voltage<br>@ 1.0A                                                           | 1.1         |     |     |     |     |     |      | V        |
| Maximum DC Reverse Current @ T <sub>A</sub> =25°C<br>at Rated DC Blocking Voltage @ T <sub>A</sub> =125°C | 5<br>50     |     |     |     |     |     |      | uA<br>uA |
| Maximum Reverse Recovery Time ( Note 1 )                                                                  | 1.8         |     |     |     |     |     |      | uS       |
| Typical Junction Capacitance ( Note 2 )                                                                   | 12          |     |     |     |     |     |      | pF       |
| Operating Temperature Range T <sub>J</sub>                                                                | -55 to +150 |     |     |     |     |     |      | °C       |
| Storage Temperature Range T <sub>STG</sub>                                                                | -55 to +150 |     |     |     |     |     |      | °C       |

Notes: 1. Reverse Recovery Test Conditions: I<sub>F</sub>=0.5A, I<sub>R</sub>=1.0A, I<sub>RR</sub>=0.25A

2. Measured at 1 MHz and Applied V<sub>R</sub>=4.0 Volts

## RATINGS AND CHARACTERISTIC CURVES (SIA THRU SIM)

FIG.1- MAXIMUM FORWARD CURRENT DERATING CURVE

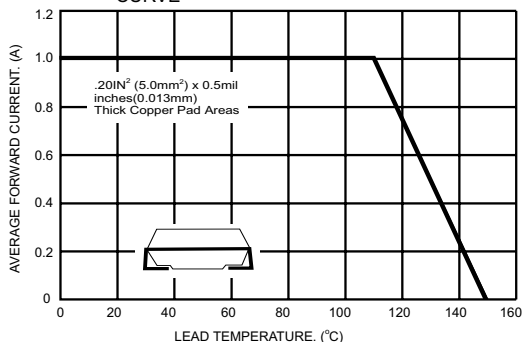


FIG.2- MAXIMUM NON-REPETITIVE FORWARD SURGE CURRENT

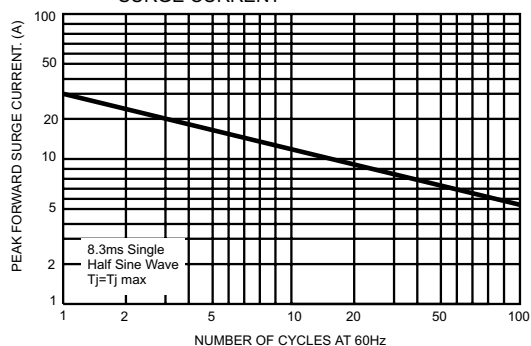


FIG.3- TYPICAL FORWARD CHARACTERISTICS

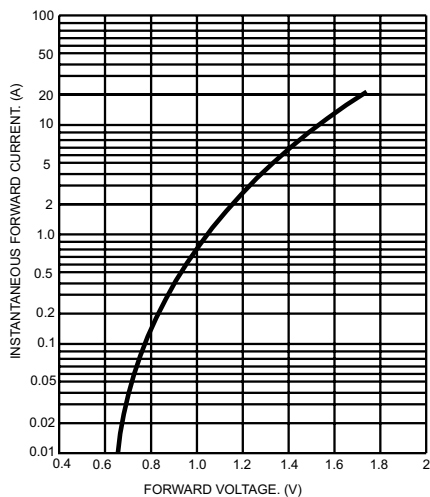


FIG.4- TYPICAL REVERSE CHARACTERISTICS

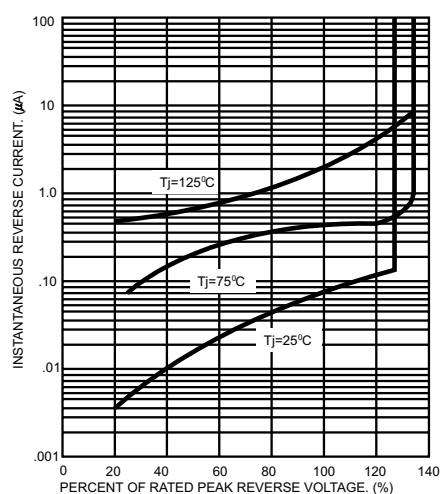


FIG.5- TYPICAL JUNCTION CAPACITANCE

